

Day 9 lab: one benchmark report for two boards

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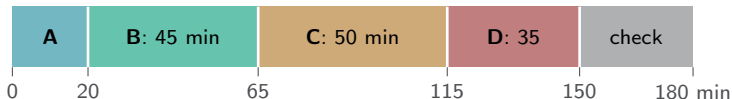
- ▶ **Goal:** your group has one benchmark report for the XIAOML Kit and the Raspberry Pi 5.
- ▶ **Boards:** the XIAOML Kit and the Raspberry Pi 5.
- ▶ **Deliverable:** the file `report.md` with the protocol, the result tables, and the Decision Log.

From the lecture to the lab

The lecture gave the method: the seven parts of a benchmark, the warm-up, the timed runs, the percentiles, the window, the energy, and the temperature. All its measured values are from an x86 computer or from other boards. Today you measure your two boards.

Key point: Each number of your report has its method: the warm-up runs, the timed runs, the input, the statistic, and the window.

Plan of the three hours



Part	Work	Time
A	Protocol: write the method of each measurement before you measure	20 min
B	Microcontroller: keyword spotting and image classification on the kit, in <code>float32</code> and in <code>int8</code>	45 min
C	Raspberry Pi: image classification and object detection, two runtimes, two thread counts, and the temperature	50 min
D	Report: latency, RAM, flash, accuracy, and estimated energy	35 min
Check by the instructor		30 min

Write each result in `report.md` when you get it.

Hardware for each group

- ▶ XIAOMI Kit and a USB-C data cable. The lab uses no sensor and no display.
- ▶ Raspberry Pi 5 (8 GB) with the active cooler, the card of your group, and the 27 W power supply.
- ▶ Laptop in the Wi-Fi edgeai-lab.
- ▶ A USB power meter, if the classroom has it.

Software

- ▶ Arduino IDE with the esp32 core 3.3.12 and the library Chirale_TensorFlowLite 2.0.0.
- ▶ On the board: `~/tflite_env` (LiteRT, ONNX Runtime) and `~/yolo` (LiteRT, NCNN).
- ▶ The model files of Days 7 and 8 on the board.
- ▶ Python 3 on the laptop for `pi/make_report.py`. It needs no package.

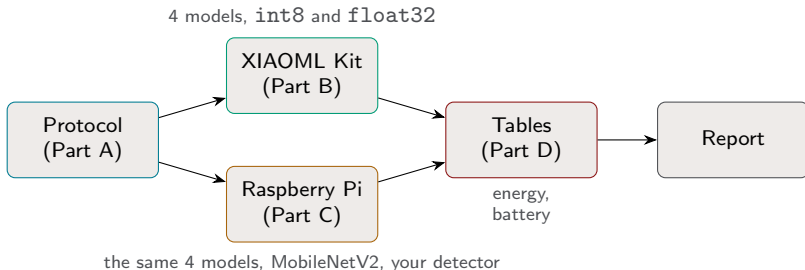
Lab files

Labs/day09/ on the laptop, `~/edgeai/day09/` on the board

The models

The four reference models of MLPerf Tiny (Part 1 of the lecture): keyword spotting and image classification

The same model file on two boards



- ▶ The kit runs the four files with TensorFlow Lite Micro. The Raspberry Pi runs the same four files with LiteRT. The two boards get the same test input.
- ▶ The Raspberry Pi also measures the models of Days 7 and 8: two runtimes and two thread counts for each model.
- ▶ The window is the model only on the two boards.

Key point: A comparison of two boards is valid only with the same model file, the same input, and the same window.

Part A: the protocol (20 min)

Step	Work	Time
1	Read the plan: three groups of measurements in the tables of Parts B and C of <code>report.md</code>	5 min
2	Fill the protocol table of Part A: the seven parts of a benchmark for the two boards	10 min
3	Write a prediction with a reason for each question of the prediction table	5 min

- ▶ Decide the warm-up runs and the timed runs. The sketch has 3 and 20 at the start, because one inference on the kit can need more than 1 second. The script of the Raspberry Pi has 20 and 200.
- ▶ The statistics: the median and the 95th percentile. The window: the model only. The input: one fixed input for each model.
- ▶ Do not change the protocol after a result. If you must change the method, write the change and the reason.

Predictions. Is `int8` faster than `float32` on the kit? How many times faster is the Raspberry Pi for the same file? Which board uses less energy for one keyword inference? Give a number and a reason for each question.

Part B: the microcontroller (45 min)

Step	Work	Time
1	Build and upload sketches/kit_bench/kit_bench.ino	10 min
2	Task B1: the function percentileUs in bench_stats.h	10 min
3	Measure the four models, copy the lines CSV, measure again	10 min
4	The clock: CPU_MHZ 80, then 240 again	10 min
5	Compare with the published results and with Day 5	5 min

- ▶ Board XIAO_ESP32S3, and Tools > PSRAM > OPI PSRAM. Set WARMUP_RUNS and TIMED_RUNS at the top of the sketch to the values of your protocol.
- ▶ The first build needs some minutes. Do Task B1 during the build: the nearest-rank rule, with integer arithmetic.
- ▶ **You record:** the two lines Sketch uses ... bytes and Global variables use ... bytes of the build.

The Serial Monitor at 115 200 baud shows Task B1: complete, and then for each of the four models:

```
--- kws int8 ---  
model: 53936 bytes  
arena: ... bytes used of 32768, in the internal RAM  
first inference: ... us  
timed runs: 20, after 3 inferences to warm up  
min ... us, median ... us, p95 ... us, max ... us  
output: class 8 (LiteRT: 8), largest difference ... Same result as LiteRT: yes  
chip temperature: ... C  
CSV,kit,kws,int8,53936,...
```

- ▶ Make the file `results_kit.csv` in `Labs/day09/`. Copy each line that starts with CSV into it.
- ▶ Same result as LiteRT: yes checks the model: the kit gives the output of the laptop for the same input.
- ▶ Send a character in the Serial Monitor: the sketch measures again. The difference of the two medians is your noise.
- ▶ **You record:** for each model the flash, the arena and its memory, the first inference, the median, the p95, and the maximum.

Step 4. Set `CPU_MHZ` to 80. Upload, and copy the new lines CSV into `results_kit.csv`, after the first lines. Then set `CPU_MHZ` to 240 again.

Step 5. The same two `int8` reference models in MLPerf Tiny v1.2:

Board	Processor	Keywords	Images
NUCLEO-L4R5ZI	Cortex-M4 at 120 MHz	43.2 ms	165.1 ms
NUCLEO-U575ZI-Q	Cortex-M33 at 160 MHz	29.1 ms	111.1 ms
NUCLEO-H7A3ZI-Q	Cortex-M7 at 280 MHz	11.3 ms	40.8 ms
Your XIAO ESP32S3	Xtensa LX7 at 240 MHz	...	...

- ▶ The three boards use an inference tool of their vendor. Your kit uses portable kernels of TensorFlow Lite Micro.
- ▶ If you have your report of Day 5, write the two times of your own keyword model. Which parts of today's protocol did Day 5 not have?

Source: MLPerf Tiny v1.2, closed division (MLCommons).
The latency is 1000 divided by the published rate.

Step	Work	Time
1	Copy the folder, connect, and check the files of Days 7 and 8	5 min
2	Task C1: the function percentile in pi/bench.py	5 min
3	Idle: the temperature and the power with no load	5 min
4	The kit models and MobileNetV2	10 min
5	The detector, in the second environment	10 min
6	A sustained run of 180 seconds	10 min
7	Copy the results to the laptop	5 min

```
scp -r day09 edge@pi-NN.local:~/edgeai/      # on the laptop, in Labs/
ssh edge@pi-NN.local                        # then on the board
cd ~/edgeai/day09
source ~/tflite_env/bin/activate
ls ../day07/models ../day08/models
python pi/bench.py --idle 30
```

- ▶ You must see `mnv2.tflite` and `mnv2_static.onnx` of Day 7, and the three `cupbottle_320` files of Day 8.
- ▶ **Task C1:** the same rule as Task B1, in Python.

```
python pi/bench.py --suite tiny           # the four models of the kit
python pi/bench.py --suite classification # MobileNetV2 of Day 7
deactivate
source ~/yolo/bin/activate                # this environment has NCNN
python pi/bench.py --suite detection       # your detector of Day 8
```

The script prints one line for each case, and adds one row to the file `results_pi.csv`:

Column	Content
task, model, runtime, type	The case: one file with one runtime
threads	1 and 4 (the suite <code>tiny</code> : 1)
first, median, p95, max	Times in ms: the first inference alone, then the timed runs
RSS MB, temp, watt	Peak RAM of the process, highest temperature, estimated power
check	<code>tiny</code> : same as the kit sketch

- ▶ Each case runs in a new process. Use `--warmup` and `--runs` for other values than 20 and 200.

```
python pi/bench.py --sustain 180 --threads 4 \  
  --model ../day08/models/cupbottle_320.tflite
```

second	inferences	median ms	p95 ms	temp C	clock MHz	throttled	watt
10	...	...	...	...	...	...	...
...							
180	...	...	...	...	...	...	...

Median latency: ... ms in the first window, ... ms in the last window (...).

- ▶ One line for each 10 seconds. The script writes the lines into `sustain.csv`.
- ▶ `throttled`: 0x0 means no event since the start of the board. Part 3 of the lecture gives the other values.
- ▶ **You record**: the first window and the last window: the median, the p95, the temperature, the clock, and the power.
- ▶ Then copy `results_pi.csv` and `sustain.csv` to the laptop with `scp` (step 7 of the README).

Prediction. Does the latency of the last window differ from the latency of the first window? Write your answer before the run.

Part D: the report (35 min)

Step	Work	Time
1	The power of each board in <code>power.csv</code>	10 min
2	Tasks D1 and D2 in <code>pi/make_report.py</code>	5 min
3	Run <code>python3 pi/make_report.py</code> on the laptop	5 min
4	The accuracy table, the answers, and the Decision Log	15 min

- ▶ **Power meter:** read the board with no load and during a run. Write the value and the method in `power.csv`.
- ▶ **No meter:** keep the data sheet values for the kit (217, 157, and 0.79 mW). For the Raspberry Pi, keep the row `pi,active` empty: the script uses the estimate of the board.
- ▶ **Task D1:** $\text{energy} = \text{power} \times \text{latency}$. **Task D2:** the mean power of a duty cycle.
- ▶ The script prints four tables and writes them into `report_tables.md`: the kit, the Raspberry Pi, the same file on the two boards, and a battery estimate.

Key point: An energy value needs its boundary and its instrument: the chip or the board, the data sheet or the meter.

Part D: the accuracy of each model

You do not measure the accuracy today. Use these values, with their source:

Model	Accuracy	Test set and source
Image classification, float32	87.2 percent	10 000 test images of CIFAR-10, this course
Image classification, int8	87.0 percent	10 000 test images of CIFAR-10, this course
Keyword spotting, int8	91.6 percent	Speech Commands, published by MLCommons
Keyword spotting, float32	not measured	models/README.md
MobileNetV2: float32, int8	77.4 and 75.2 percent	500 images of Imagenette, the Day 7 lab
Your detector	your values	Your notebook of Day 8

- ▶ The two image files give the same accuracy on the 200 test images of MLPerf Tiny: 87.0 percent. The interval of 200 images is plus and minus 4.7 points.
- ▶ A latency with no accuracy can reward the wrong file. Each row of your report needs the two numbers.

Source: Image classification: an experiment of this course.

Keyword spotting: Banbury et al., 2021.

The instructor checks this at the end of the lab:

- ▶ ✓ Part A has the protocol and the predictions, and no prediction looks like a result.
- ▶ ✓ The Serial Monitor shows Task B1: complete, and Same result as LiteRT: yes for each of the four models.
- ▶ ✓ `python pi/bench.py --suite tiny` prints Task C1: complete and same as the kit sketch for the four models.
- ▶ ✓ `python3 pi/make_report.py` prints Tasks D1 and D2: complete.
- ▶ ✓ Each number of the report has a method: the warm-up runs, the timed runs, the input, the statistic, and the window.
- ▶ ✓ The report compares the same model file on the two boards: the latency and the energy.
- ▶ ✓ The report has the temperature of the sustained run.
- ▶ ✓ The Decision Log gives numbers and names one trade-off.

Hand in: the file `report.md`, with the four tables of `report_tables.md`.

Write about 100 words. State one design decision, give your measured numbers, and name the trade-off.

A company wants a keyword detector that runs from a battery for one month, and a camera that counts cups on a desk with 10 frames in each second.

- ▶ Which board do you select for each product?
- ▶ Which precision, which runtime, and how many threads?

A good Decision Log:

- ▶ gives each number with its method: the runs, the window, and the instrument of the power,
- ▶ uses the battery table of Part D for the keyword detector,
- ▶ names what you lose with your decision. Example: the kit cannot run the detector fast enough, and the Raspberry Pi cannot run for one month from a small battery.

If a part does not work, and common problems

Problem	Fix or fallback
ERROR: no memory for an arena	Set OPI PSRAM and upload again. If the error stays, write “does not fit”: this is also a result.
The median and the p95 of the kit are 0	Task B1 is not complete
Same result as LiteRT: NO	Copy the sketch folder again. Write the library version in the report.
The Serial Monitor shows nothing	Send a character: the sketch measures again
No module named 'ncnn'	deactivate, then source ~/yolo/bin/activate
pi/bench.py prints not found	Copy the file of Day 7 or Day 8 with scp, or give it with --model
The column watt is empty	Write your meter value, or the published range of 5 to 7 W, in the row pi, active of power.csv
The throttle state is not 0x0	Use the 27 W supply, check the cooler, and write the value in the report
A task is not complete in time	Use the file of solutions/, and write this in the report

The README of the lab has the complete list.

This lecture adapts material from these sources:

- ▶ *Machine Learning Systems* by Vijay Janapa Reddi and contributors, Harvard University. mlsysbook.ai, CC BY-NC-SA 4.0.
- ▶ *Edge AI Engineering: Raspberry Pi* by Marcelo Rovai. github.com/Mjrovai/EdgeML_Made_Ease_ebook and github.com/Mjrovai/EdgeML-with-Raspberry-Pi, GPL-3.0.

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